

Mechanochemical Feedback Drives Complex Inertial Dynamics in Active Solids

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Active solids combine internal active driving with elasticity to realize states with nonequilibrium mechanics and autonomous motion. They are often studied in overdamped settings, e.g., in soft materials, and the role of inertia is less explored. We construct a model of a chemically active solid that incorporates mechanochemical feedback and show that, when feedback overwhelms mechanical damping, autonomous inertial dynamics can spontaneously emerge through sustained consumption of chemical fuel. By combining numerical simulations, analysis, and dynamical systems approaches, we show how active feedback drives complex nonlinear dynamics on multiple timescales, including limit cycles and chaos. Our results suggest design principles for creating ultrafast actuators and autonomous machines from soft, chemically powered solids.

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Introduction—Soft materials driven by chemical stimuli have been widely used as actuators to achieve complex tasks—locomotion, grasping, and shape morphing—through flexible deformations [1,2]. While promising for technologies [3], their contraction speed and power output are often limited by slow, diffusive mechanisms (thermal, fluidic, ionic, etc.) and damping [4,5]. To overcome this limitation, recent approaches have used miniaturization [6,7] or inertial dynamics triggered by plant-inspired elastic instabilities [8–10] and fast chemical processes (e.g., combustion) [11–14] to achieve fast actuation. However, such systems passively relax without sustained input, prompting us to ask: How can soft, chemically driven materials sustain fast and continuous inertial motion without cyclic stimulation, and how complex can this motion be?

A potential strategy is suggested by contractile tissues that use biological feedback to drive extreme motions in animals [15,16]. Such active solids, exemplified by living tissues [16–19], cellular and organismal collectives [20,21], mechanochemical gels [22–24], and robotic assemblies [25], have been shown to produce exotic dynamics [16,18–20,26–29] and present new opportunities for designing soft chemically driven machines.

Building upon previous studies on biological systems with overdamped feedback [16,19,27,30–33] and inertial robotic assemblies [34–36], we combine both effects to demonstrate how chemically active soft solids with sufficiently strong mechanochemical feedback can sustain complex and autonomous nonlinear inertial dynamics. We show how feedback introduces new timescales that govern how stress accelerates or decelerates chemical reactions. When this response becomes faster than viscous dissipation of momentum, the system continually extracts chemical energy to power self-sustained inertial oscillations and chaotic dynamics. This suggests mechanochemical feedback as a promising scalable mechanism for devising soft engines, actuators, and autonomous machines with fast and rich dynamics.

Mechanochemical active solid—We model the active solid as a one-dimensional (1D) gel of length L and mass density ρ suffused with a mixture of chemical species A and B (concentration n_A, n_B) that interconvert into each other via a reversible reaction $A \rightleftharpoons B$. The total concentration is conserved by the reaction ($n_A + n_B = n_0$), so the relative concentration difference defined by $\chi = (n_B - n_A)/n_0$ serves as a convenient reaction coordinate. Material deformations are assumed small and captured by a linearized strain $\epsilon(x, t) = \partial_x u(x, t)$ (u , displacement; x , material coordinate). The free energy of the system [37–40],

$$\mathcal{F} = k_B T \int dx \left[n_0 f(\chi) + \frac{E}{2k_B T} \epsilon^2 + n_0 C \chi \epsilon \right], \quad (1)$$

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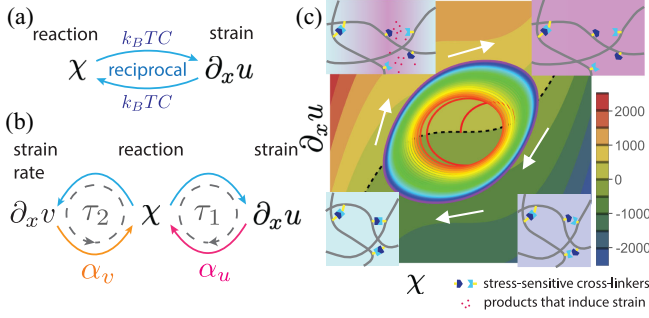


FIG. 1. Mechanochemical coupling. (a),(b) Feedback loops among reaction (χ), strain ($\partial_x u$), strain rate ($\partial_x v$) in passive (a) and active (b) solids. In the passive limit (a), detailed balance enforces vanishing strain-rate coupling and a reciprocal strain-reaction coupling ($k_B TC$). When active (b), the feedback drives nonequilibrium fluxes on independent timescales $\tau_{1,2}$. (c) Contour plot of reaction rate $\partial_t \chi$ (in A.U.) using the 0D model in Eq. (5), with the dashed line marking $\partial_t \chi = 0$ so χ grows (decays) above (below) this curve. An example trajectory is shown where a perturbation from the origin leads to a limit cycle, with four representative states illustrated as insets, where the background shading refers to concentration χ . Starting with the bottom left corner, increased strain breaks stress-sensitive cross-linkers, which drives χ to increase, followed by decrease in strain, and recovery back to the first state.

combines a local free energy density $f(\chi)$, an elastic energy with modulus E , and a passive coupling C between the chemical concentration and deformations, with T as the temperature. The passive coupling controls the chemically induced mechanical stresses present, e.g., as can be seen in chemically responsive gels [22,24], solvent-driven swelling [9], and explosives or combustion [14,41].

The mechanical stress in the material $\sigma = \sigma^p + \sigma^d$ consists of both passive ($\sigma^p = \delta \mathcal{F} / \delta \epsilon = E\epsilon + n_0 k_B TC \chi$) and dissipative ($\sigma^d = \eta \partial_t \epsilon$) contributions [38–40], with η a viscosity. In the absence of any reactive chemicals ($n_0 = 0$), the material behaves as a passive Kelvin-Voigt viscoelastic solid. Kinematics and momentum conservation then give

$$\begin{aligned} \partial_t u &= v, \\ \rho \partial_t v &= \partial_x (\sigma^p + \sigma^d), \end{aligned} \quad (2)$$

where $v(x, t)$ is the local material velocity.

Chemical reactions involve transitions between molecular states and, in equilibrium, a balance of forward and backward transition rates [42,43]. That mechanical deformations can modify reaction rates (kinetics) and bias chemical equilibria (energetics) is known in diverse systems, e.g., mechanochemically responsive polymers [44–46], inorganic compounds [47,48], biomolecules [49–51], and living cells [52,53].

Following standard transition state theory [42,54] generalized for nonideal systems [55,56], we have forward and

backward reaction rates given by $r_+ = k_0 \exp(\mu_A / k_B T)$ and $r_- = k_0 \exp(\mu_B / k_B T)$, respectively, where μ_A, μ_B are the chemical potentials. Here k_0 is a microscopic rate constant that depends on details of the transition state. In ideal systems, these rates reduce to the law of mass action. In a passive system, $\mu_i^p = \delta \mathcal{F} / \delta n_i$, so $\mu_B^p = -\mu_A^p = k_B T [f'(\chi) + C\epsilon]$. This relation demonstrates reciprocity between the reaction and strain ($\delta \mu_i^p / \delta \epsilon = \delta \sigma^p / \delta n_i = \mp k_B TC$, $i = A, B$) [37,43] and captures the energetic shift in chemical equilibria due to deformations.

To drive the system out of chemical equilibrium, we augment the reaction rates by setting $\mu_i = \mu_i^p + \mu_i^a$, where an external source of reactive free energy, say, from an abundant supply of a fuel that catalyses the reaction [39,40], drives a bias in chemical potential. A constant bias simply changes the equilibrium concentration of reactants and products by a fixed amount and leaves the system passive, but a dynamic bias can induce an active coupling between mechanical stress and the reaction beyond the free energy in Eq. (1). In particular, mechanical stress can modify both the transition state barrier and the chemical potential difference, which we minimally capture by setting $\mu_i^a = h_i \sigma$, where σ is the total stress and h_i are mechanochemical coupling constants. This feedback breaks reciprocity as $\delta \mu_i / \delta \epsilon \neq \delta \sigma^p / \delta n_i$ and provides a nonequilibrium coupling between the reaction and mechanical degrees of freedom.

Collecting these effects together, and assuming $\mu_i \ll k_B T$ (the chemical potential associated with the reaction is small) to linearize the reaction rates, we obtain (see more details in Supplemental Material [57])

$$\partial_t \chi = r_+ - r_- = r_{\text{eq}} + \zeta \sigma, \quad (3)$$

where $r_{\text{eq}} = (k_0 / k_B T)(\mu_A^p - \mu_B^p) = -2k_0 [f'(\chi) + C\epsilon]$ is the equilibrium reaction rate including the passive coupling to strain and $\zeta = k_0(h_A - h_B) / k_B T$ is the nonequilibrium coupling to stress. Note that, for the system to be passive, we need $\zeta = 0$, which invalidates coupling to strain rate. Equations (2) and (3), along with initial and boundary conditions, completely specify the inertial active solid model.

1D model—To exemplify its key consequences, we consider a bistable chemical free energy $f(\chi) = -a_2 \chi^2 / 2 + a_4 \chi^4 / 4$ with $a_2, a_4 > 0$. The model [Eqs. (2) and (3)] then simplifies to give

$$\begin{aligned} \partial_t u &= v, \\ \rho \partial_t v &= E \partial_x^2 u + \eta \partial_x^2 v + c \partial_x \chi, \\ \tau_c \partial_t \chi &= \chi(1 - g\chi^2) - \alpha_u \partial_x u - \alpha_v \partial_x v, \end{aligned} \quad (4)$$

where $c = n_0 k_B TC$ is the passive chemically induced stress scale, $\tau_c = 1 / [2k_0 a_2 + \zeta c]$ is the effective chemical reaction timescale, combining the equilibrium rate ($2k_0 a_2$) and

the active feedback rate (ζc) [57], $g = 2k_0 a_4 \tau_c$ controls the chemical nonlinearity, and $\alpha_u = \tau_c(2k_0 C - \zeta E)$ and $\alpha_v = -\tau_c \zeta \eta$ are the nonequilibrium feedback couplings to strain and strain rate, respectively [Figs. 1(a) and 1(b)].

There are three homogeneous steady-state solutions, the trivial one ($\chi = \chi_{ss}, \epsilon = \epsilon_{ss}$, and $v = 0$), which always exists, and two nontrivial ones [$\epsilon_{ss} = \pm(c/E)\sqrt{(E + c\alpha_u)/Eg}$, $\chi_{ss} = \mp\sqrt{(E + c\alpha_u)/Eg}$, and $v = 0$], which only exist for $c\alpha_u < -E$. We numerically solve Eq. (4) in a 1D domain of length $L = 2\pi$ with periodic boundary conditions using the pseudospectral method [58]. We pick time, length, mass, and reaction coordinate scales so that $\rho = E = c = g = 1$ and vary the activity parameters, α_u and α_v .

In Fig. 2, we showcase rich dynamics that can emerge from this 1D model. For $\alpha_v = 0$ (only strain feeding back to reaction), as α_u decreases from positive (reciprocal) to negative (nonreciprocal), the system changes from a phase separated steady state [Fig. 2(a)] to a homogeneous steady state when $\alpha_u < -E/c$ [Fig. 2(b)] and develops a limit cycle (LC) at large negative α_u [Fig. 2(c)]. On the other hand, fixing $\alpha_u > 0$ and increasing α_v (strain rate also feeds back to reaction) triggers nonequilibrium states with increasing complexity: LC [Fig. 2(d)], period doubling [Fig. 2(e)], and chaos [Fig. 2(f)]. These dynamical states caused by $\alpha_u < 0$ or $\alpha_v > 0$ are sustained by continuous conversion of chemical energy to mechanical energy. While nonreciprocal couplings are known to be a generic mechanism for creating dynamic patterns in systems with conservation laws [26,59–61], here we focus on their interplay with inertia.

Scaling analysis reveals the key features underlying the transitions between the qualitative behaviors exhibited in Fig. 2. In the absence of a chemical reaction ($\chi = 0$), long-wavelength deformations on the scale of the system size L transport momentum by elastic waves on an inertial timescale $\tau_{el} \sim L\sqrt{\rho/E}$ and dissipate it through viscous stresses

on a diffusive (Stokes) timescale $\tau_\eta \sim \rho L^2/\eta$. Importantly, we note that τ_η depends on inertia and is distinct from the local viscoelastic relaxation time $\tau_{ve} \sim \eta/E$. The bare reaction kinetics, on the other hand, proceeds on a chemical timescale $\sim \tau_c$, but nonequilibrium mechanochemical feedback can pump energy into the system at independent rates. To see this, we consider the time taken for a minimal feedback cycle [Figs. 1(b) and 1(c)]. A chemically induced stress $\sim c\chi$ generates a characteristic elastic strain $|\epsilon| \sim c\chi/E$ or viscous strain rate $|\dot{\epsilon}| \sim c\chi/\eta$ that can, in turn, drive the reaction via $\tau_c \dot{\chi} \sim \alpha_u \epsilon$ or $\tau_c \dot{\chi} \sim \alpha_v |\dot{\epsilon}|$, completing the cycle. This gives two new timescales, $\tau_1 \sim \tau_c E/|\alpha_u c|$ and $\tau_2 \sim \tau_c \eta/|\alpha_v c|$, set by the strain and strain-rate feedback terms, respectively [Fig. 1(b)]. When activity overwhelms the rate of mechanical dissipation, i.e., $\tau_1, \tau_2 \lesssim \tau_\eta, \tau_{ve}$, we expect complex dynamics to emerge (Fig. 2). Note that for the examples in Fig. 2 we choose $\eta = 0.16$, so $\tau_\eta > \tau_{el}$ and the system is mechanically underdamped.

Effective 0D model—We next perform a single mode Galerkin approximation to reduce the spatial complexity of the 1D model by projecting Eq. (4) onto the first fundamental mode of the 1D system. We set $u(x, t) = (1/q)U(t) \sin(qx)$, $v(x, t) = (\sqrt{E/\rho})V(t) \sin(qx)$, and $\chi(x, t) = (E/c)X(t) \cos(qx)$, with $q = 2\pi/L$ to nondimensionalize the equations and obtain

$$\begin{aligned} \tau_{el} \dot{U} &= V, \\ \tau_{el} \dot{V} &= -U - \text{Vi}V - X, \\ \tau_{el} \text{Ch} \dot{X} &= X(1 - X^2) - A_u U - A_v V, \end{aligned} \quad (5)$$

where we have assumed $g = (4/3)(c/E)^2$ to simplify the chemical nonlinearity and set $\tau_{el} = (L/2\pi)\sqrt{\rho/E}$ as the inertial time for a long-wavelength elastic wave. The relative importance of viscous dissipation to inertia defines

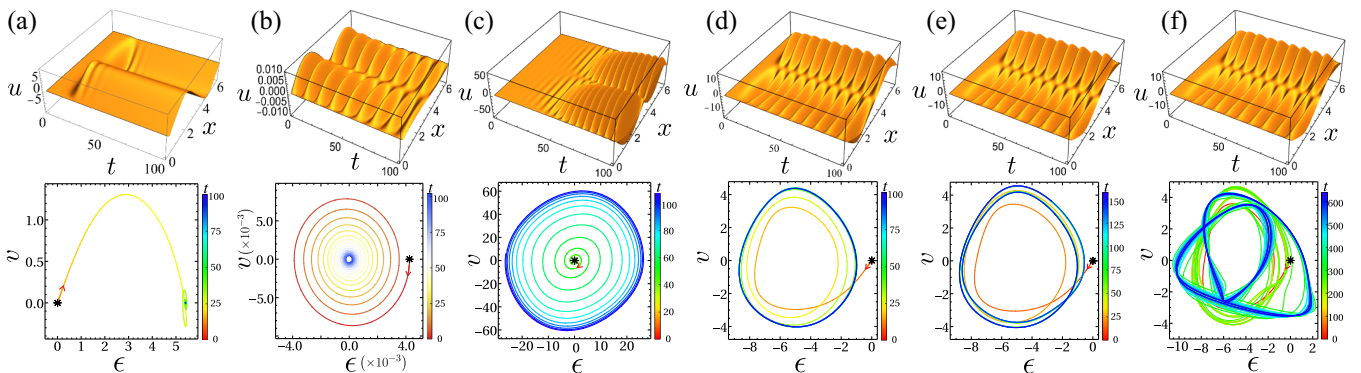


FIG. 2. Dynamics of the 1D mechanochemical system in Eq. (4). We set $L = 2\pi$, $\eta = 0.16$, and $\tau_c = 100$ along with (a) $\alpha_u = 50$, $\alpha_v = 0$, (b) $\alpha_u = -10$, $\alpha_v = 0$, (c) $\alpha_u = -50$, $\alpha_v = 0$, (d) $\alpha_u = 50$, $\alpha_v = 77$, (e) $\alpha_u = 50$, $\alpha_v = 80$, (f) $\alpha_u = 50$, $\alpha_v = 82$. We include a small diffusion constant $D = 0.05$ to regularize chemical gradients. The initial conditions are $u(x, 0) = 0.01 \sin(x)$, $v(x, 0) = 0 = \chi(x, 0)$. The first row shows the displacement field $u(x, t)$. The second row shows the phase space ($\epsilon = \partial_x u, v$) trajectory at the point $x = 9\pi/25$.

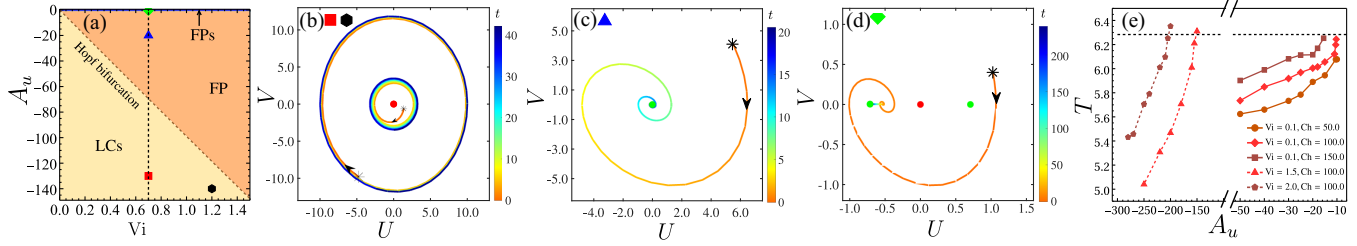


FIG. 3. Dynamical phases due to nonreciprocal strain-chemistry feedback ($A_v = 0$, $A_u < 0$) in the 0D model [Eq. (5)]. (a) Phase diagram ($Ch = 100$). (b)–(d) Typical trajectories in the $U - V$ plane for (b) the outer (larger) LC with $(V_i, A_u) = (0.70, -130.0)$ (underdamped regime) and the inner (smaller) LC with $(V_i, A_u) = (1.2, -140.0)$ (overdamped regime), (c) one FP at $U_0 = X_0 = 0$, $V_0 = 0$ with $(V_i, A_u) = (0.70, -20.0)$, and (d) FPs with the trajectory converging to one of them $U_+ = -X_+ = \sqrt{1 + A_u}$ with $(V_i, A_u) = (0.70, -0.5)$. The FPs are shown as red (unstable) and green (stable) dots in (b)–(d), with the arbitrarily chosen initial conditions denoted by the black star. In (b)–(d), different symbols adjacent to the panel index represent the corresponding points in the phase diagram in (a). (e) Dependence of the time period T on A_u for different Ch in the LC region (in $\tau_{el} = 1$ units).

a viscoinertial number $Vi = 2\pi\eta/(L\sqrt{\rho E}) \sim \tau_{el}/\tau_\eta$. Similarly, the ratio of the chemical reaction time to that of inertia defines a chemoinertial number $Ch = (2\pi\tau_c/L)\sqrt{E/\rho} = \tau_c/\tau_{el}$. As we are primarily interested in the mechanically underdamped regime with slow chemical reactions, we expect $\tau_{el} < \tau_\eta < \tau_c$, i.e., $Vi < 1$ and $Ch \gg 1$. Finally, nonequilibrium activity is captured by two feedback parameters: $A_u = \alpha_u c/E \sim \tau_c/\tau_1$ and $A_v = 2\pi\alpha_v c/(L\sqrt{\rho E}) \sim Vi\tau_c/\tau_2$. Note that, when $A_u < 0$, the mechanochemical coupling is nonreciprocal. With initial conditions, the four nondimensional parameters Vi , Ch , A_u , and A_v completely define the response of our 0D system. Mathematically similar models appear in quantum and classical control systems [62–67], metamaterials [68], and in the physics of hearing [69,70], but the characterization of the nonlinear dynamics in the context of inertial mechanochemical solids is novel to our knowledge. As in the 1D model, for steady state, Eq. (5) have (i) one fixed point (FP) $U_0 = X_0 = 0$ for $A_u < -1$ and (ii) three FPs $U_\pm = -X_\pm = \pm\sqrt{1 + A_u}$ and $U_0 = X_0 = 0$ for $A_u > -1$ (at FPs, by definition, $V = 0$).

At $A_v = 0$, nonreciprocity between strain and reaction arises for $A_u < 0$. Upon decreasing A_u , a linear stability analysis reveals first a pitchfork bifurcation where three FPs (bistable) transition to one FP (monostable) and then an oscillatory Hopf bifurcation occurs at a second threshold when $A_u \approx -ViCh$ [71] (Fig. 3). Figure 3 shows typical trajectories in the $U - V$ plane for limit cycles (b) and FPs (c),(d). The analytical bifurcation thresholds (derived in Supplemental Material [57]) are confirmed via numerical continuation using MatCont and shown in Fig. 3(a). This Hopf bifurcation corresponds to $\tau_1 \lesssim \tau_\eta$, i.e., when the strain feedback actively pumps energy faster than viscous stresses can dissipate it, and it crucially depends on inertia as $\tau_\eta \sim \rho L^2/\eta$.

Near the Hopf bifurcation, the imaginary part of the unstable eigenvalue yields the time period of the LC to be $T = 2\pi\tau_{el}/\sqrt{1 - (Vi/Ch)} \approx 2\pi\tau_{el}$ for $Ch \gg 1$ [dashed

line in Fig. 3(e)]: fast oscillations at the inertial timescale, distinct from previous examples of spontaneous oscillations in overdamped active solids [16,17,19,27,72]. Interestingly, this instability and LC behavior persists even in the mechanically overdamped regime ($Vi > 1$, see Supplemental Material [57] for more details) with a period controlled by the same τ_{el} , as long as nonreciprocal feedback overwhelms the mechanical dissipation rate [Figs. 3(b) and 3(e)].

Now we consider $A_u > 0$, $A_v > 0$, where strain rate steers the activity (Fig. 4). By choosing $A_u > 0$, we have three FPs, with the FP at the origin (U_0, X_0) always unstable. A linear stability analysis shows that, upon increasing A_v , both nontrivial FPs ($U_\pm, X_\pm$) lose stability via a Hopf bifurcation when $A_v \gtrsim Ch$ (valid in the limit of slow kinetics, i.e., $Ch \gg Vi, A_u$, see Supplemental Material [57] for the full expression). This analytical bifurcation threshold [white dashed line in Fig. 4(a)] is confirmed by numerical analysis. In terms of physical timescales, this bifurcation condition corresponds to $\tau_2 < \tau_{ve}$, i.e., the strain-rate feedback time is smaller than the local viscoelastic relaxation time. Near onset, the small LC has a slow oscillation time period $T_{\text{small}} \approx \pi\tau_{el}\sqrt{2ViCh/(1 + A_u)} \sim \sqrt{\eta\tau_c/E}$ governed by dissipative processes. Note that these timescales do not rely on inertia, and the resulting instability is smoothly connected to previously studied instabilities in overdamped active solids [16,17,19,27,72].

Enabled by the three dimensionality of phase space (U , V , X) including inertia, at large A_v , chaos arises. We find a period doubling [Fig. 4(d)] cascade for the pair of LCs, which collide into a chaotic strange attractor [Fig. 4(e)], reminiscent of the classical Lorenz and Rossler system [73]. Chaos develops in a large parameter regime in the $A_v - Vi$ space [Fig. 4(a)] and is interspersed by small islands of periodic orbits [Fig. 4(g) and Supplemental Material [57] for Lyapunov exponent analysis]. The phase boundaries separating chaos from limit cycles in Fig. 4(a)

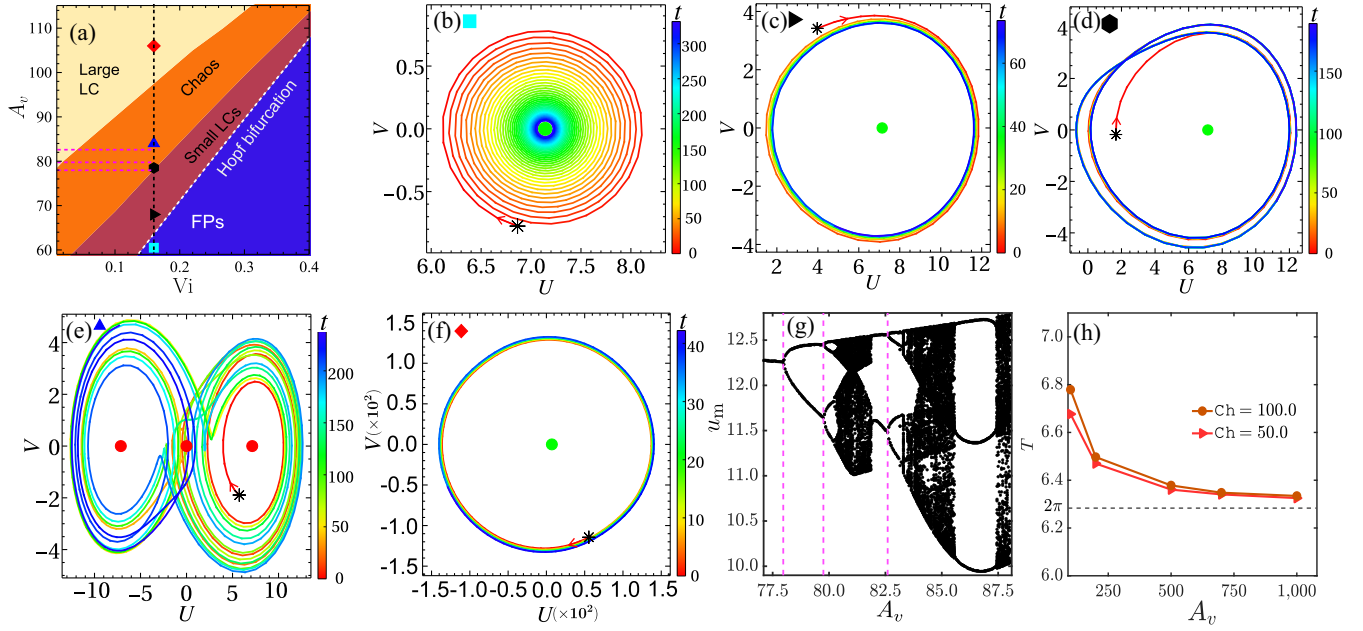


FIG. 4. Dynamical phases due to nonreciprocal strain-strain rate chemistry feedback ($A_v > 0$, $A_u > 0$) in the 0D model [Eq. (5)]. (a) Phase diagram ($\text{Ch} = 100$, $A_u = 50$). (b)–(f) Typical trajectories in the $U - V$ plane for the FPs with the trajectory converging to (b) one FP $U_+ = -X_+ = \sqrt{1 + A_u}$ for $A_v = 60.5$, (c) an LC around one unstable FP for $A_v = 68.0$, (d) period doubling for $A_v = 78.5$, (e) chaos for $A_v = 84.0$, and (f) the large LC enclosing all FPs for $A_v = 106.0$. In all cases, we set $V_i = 0.16$ and $A_u = 50.0$. The black stars in (b)–(f) denote the arbitrarily chosen initial condition. (g) Bifurcation diagram showing the route from LC around the FP $U_{\pm} = -X_{\pm} = \sqrt{1 + A_u}$ to chaos as a function of the parameter A_v . Here, u_m represents the local maximum value of U [57]. (h) Dependence of the time period T on A_v at $V_i = 0.16$ in the large LC region, approaching 2π as A_v increases (in $\tau_{el} = 1$ units).

are obtained by analyzing phase space trajectories obtained by direct numerical simulations. For even larger values of A_v , eventually, the system settles into a globally attracting large LC that encloses all three FPs [Fig. 4(f)]. While the small LCs have slow oscillations, we numerically find that the large LC exhibits fast inertial oscillations with a time period approaching $T \approx 2\pi\tau_{el}$ as $A_v \rightarrow \infty$ [Fig. 4(h)]. This explains and recapitulates the complex behavior observed in the 1D model (Fig. 2). Whether such chaotic and fast dynamics observed for $A_v > 0$ survives when $V_i \gtrsim 1$ is left for future work.

Conclusion—By using nonequilibrium feedback to couple chemical reactions with the mechanics of a soft solid, we have shown how active driving can sustain autonomous deformations, fast oscillations, and complex dynamics. These results highlight the importance of inertia in active mechanochemical solids and provide a design principle for generating high quality ultrafast oscillators even in dissipative, soft solids. That feedback can also trigger chaos in active solids suggests a simple way to generate complex “twitching” movements, reminiscent of rudimentary behaviors in soft-bodied organisms [15,74]. Our Letter paves the way to explore a new regime of chemically active solids in previously inaccessible high frequency scales and provides a scalable and efficient platform to design autonomous soft materials from homogeneous media, without the need for complex microstructures or discretely embedded active particles.

Recent experiments suggest a route to test our predictions. A promising platform is pH-sensitive hydrogels that can achieve reversible swelling and actuation with $c/E \sim 0.1\text{--}0.5\text{ pH}^{-1}$ (assuming $\chi \sim \text{pH}$) [22,24,75]. Rapid feedback can be engineered, e.g., using swelling induced mechanophore activation [76] or strain triggered acidification [77] that occurs within a second, providing an estimate of $|\alpha_u|/\tau_c \sim 0.1\text{ pH/s}$. When combined in a single material, this would result in $\tau_1 \sim 10^{-2}\text{ s}$ [57]. To observe the predicted inertial Hopf bifurcation, we need $\tau_1 < \tau_\eta$ which requires a viscosity $\eta \sim 10\text{--}100\text{ Pa s}$ (assuming $\rho \sim 10^3\text{ kg/m}^3$ and $L \sim 1\text{--}3\text{ cm}$), potentially achievable in low-dissipation hydrogels [78]. Typical values of small molecule diffusion constants in hydrogels, $D \sim 10^{-9}\text{ m}^2/\text{s}$ [79], yield a diffusion timescale $L^2/D \sim 10^5\text{ s}$, much larger than timescales of interest in this problem, justifying our neglect of molecular diffusion. Similar estimates can be made for strain-rate feedback that has been qualitatively observed, e.g., in Ref. [77]. Biological materials, e.g., actomyosin or muscle tissue, can achieve strong active stresses and feedback with $c/E \sim 0.1\text{--}1$, $\tau_c \sim 10^{-3}\text{ s}$, $|\alpha_u| \sim 0.1\text{--}10$, and $|\alpha_v|/\tau_c \sim 1\text{--}10$ (using $\chi \sim$ myosin fraction) [16,51], leading to $\tau_1 \sim 10^{-4}\text{--}0.1\text{ s}$ and $\tau_2/\tau_{ve} \sim 0.1\text{--}10$. But their large viscosity $\eta \gtrsim 10^5\text{ Pa s}$ [16,80] suppresses the inertial instability requiring $\tau_1 < \tau_\eta \lesssim 10^{-8}\text{ s}$ (assuming $\rho \sim 10^3\text{ kg/m}^3$ and

$L \sim 1$ mm), though $\tau_2 < \tau_{ve}$ permits the overdamped small LC to exist [16,17,27]. This suggests combining weak dissipation with strong feedback as a simple principle for designing inertial mechanochemical metamaterials, powered by and consisting of collectives of, e.g., chemically fueled microrobots [81].

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Data availability—The data that support the findings of this article are openly available [82].

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